



IN-SEMESTER EXAMINATION – 2025-26 (Semester-I)

Date: Friday, 22-Aug-2025

Course: Bridge Analysis and Design

Time: 11:00 – 11:45

Class: Final Year Civil Engineering

Course Code: PEC-CE412

Maximum Marks: 20

Note 1. Attempt all questions.

2. Use of non-programmable calculator is permitted.

3. Draw appropriate figures wherever required.

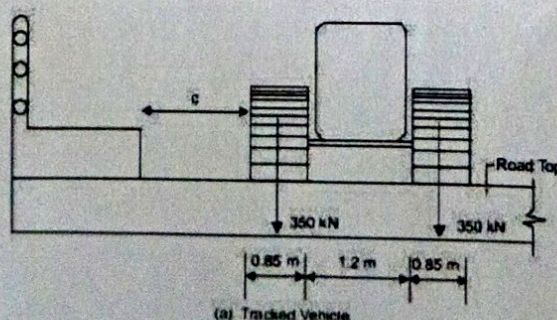
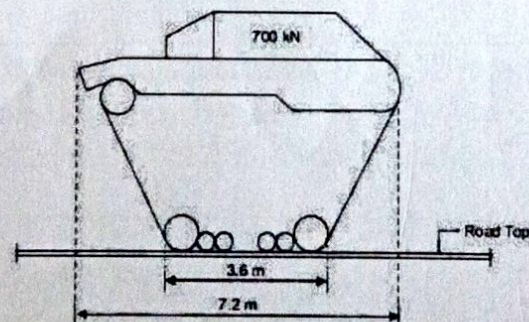
4. Assume appropriate data, if necessary.

Que. No.	Question statement	Mark	CO	BT															
01	Determine the load intensity on a slab having total width of 8.7 m and span length of 5.37 m. The kerbs on both sides having width of 600 mm. The thickness is 56 mm and 420 mm of wearing coat and slab, respectively. Also, determine the maximum bending moment along span.	10	CO2	BT4															
02	Determine the economical span for the bridge. The cost of one substructure and one superstructure a span of multiple span bridge for various span lengths is presented in following table. <table><tr><td>Span (m)</td><td>4</td><td>8</td><td>12</td><td>15</td></tr><tr><td>Superstructure cost</td><td>1600.00</td><td>7500.00</td><td>16800.00</td><td>24550.00</td></tr><tr><td>Substructure cost</td><td>22200.00</td><td>23200.00</td><td>23000.00</td><td>23600.00</td></tr></table>	Span (m)	4	8	12	15	Superstructure cost	1600.00	7500.00	16800.00	24550.00	Substructure cost	22200.00	23200.00	23000.00	23600.00	05	CO2	BT3
Span (m)	4	8	12	15															
Superstructure cost	1600.00	7500.00	16800.00	24550.00															
Substructure cost	22200.00	23200.00	23000.00	23600.00															
03	Write a short note on IRC class A load.	05	CO1	BT2															

Course Outcome (CO)	Bloom's Taxonomy Level (BT)
CO1 To recall the fundamentals of water resources engineering.	BT1 – Remember
CO2 To understand the components of bridges, its classifications and importance.	BT2 – Understand
CO3 To apply various theories of lateral load distribution for component design of bridge.	BT3 – Apply
CO4 To analyse any given type of bridge.	BT4 – Analyse
CO5 To get exposure to evaluation of sub structures, type of foundations, importance of bearings, lessons from bridge failures.	BT5 – Evaluate
	BT6 – Create

*** All the Very Best ***

DATA REQUIRED AS PER THE RELEVANT IRC codes:



(where, $c = 1.20$ m)

b/l_e	α for simply supported slab	α for continuous slab	b/l_e	α for simply supported slab	α for continuous slab
0.1	0.40	0.40	1.1	2.60	2.28
0.2	0.80	0.80	1.2	2.64	2.36
0.3	1.16	1.16	1.3	2.72	2.40
0.4	1.48	1.44	1.4	2.80	2.48
0.5	1.72	1.68	1.5	2.84	2.48
0.6	1.96	1.84	1.6	2.88	2.52
0.7	2.12	1.96	1.7	2.92	2.56
0.8	2.24	2.08	1.8	2.96	2.60
0.9	2.36	2.16	1.9	3.00	2.60
1.0	2.48	2.24	2 & above	3.00	2.60



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded

Department of Civil Engineering
In Semester Examination-I (ISE-I)
Academic Year: 2025-26 (Semester I)

Class: Final Year B.Tech	Subject: Foundation Engineering	Subject Code: PCC-CE403
Date: 19.08.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.

Q.N.	Questions	Marks	CO	BT
Q.1	Explain in detail the standard penetration test of soil with neat sketch along with its corrections.	05	1	1
Q.2	Explain in details about auger boring in soil.	05	1	1
Q.3	Sub-surface exploration at the site of a proposed building reveals the existence of a 2.4 metre thick layer of soft clay below a stratum of coarse sand which is 4 metre thick and extends from the ground surface upto the top of the clay layer. The ground water table is at 2.5 metre below the ground surface. Laboratory tests indicate the natural water content of clay as 40%, liquid limit 45% and specific gravity 2.75. The unit weights of sand above and below the water table are respectively 17.8 kN/m^3 and 21.0 kN/m^3 . Estimate the probable final settlement of the building, if its construction will increase the average vertical pressure on the clay layer by 71 kN/m^2 .	07	1	3
Q.4	A test plate 30 cm square settles by 12 mm under a load of 4.5 kN in a sandy soil. By how much will a footing 2 m x 2 m subjected to a load of 200 kN settle in mm?	03	2	3

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Recall the fundamentals of building construction and geotechnical engineering.
CO 2	Identifying the applicability of different foundation design applicable to various types of soil.
CO 3	Apply the different design philosophy of foundation applicable to soil.
CO 4	Analysis various types of shallow and deep foundations.
CO 5	Create understanding applicability of foundation design for different types of soil.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded

ISE-I (Semester-I) 2025-26

Final Year B. Tech.

Civil Engineering

Course: Engineering Economics, Estimation and Costing

Course Code: PCC-CE402

Date: 21-Aug-2025

Time: 45 Minutes (11:00 to 11:45)

Max. Marks: 20

Q. No.	Questions (Solve any FOUR)	Marks	CO	BT
Q. 1)	If the price of certain goods falls from Rs. 20 to Rs. 10, that causes an increase in the demand from 43 units to 75 units. Calculate the price elasticity of demand and determine the type of price elasticity.	05	2	4
Q. 2)	Show the graphical representation of demand and supply curve. Write down the uses of demand and supply curve.	05	2	2
Q. 3)	Write down the advantages and disadvantages of indirect tax.	05	2	2
Q. 4)	What is life cycle cost? Explain it with an example.	05	2	2
Q. 5)	What is break-even analysis? Write down its importances.	05	2	2
Q. 6)	Rs. 1,000 invested for five years in an investment scheme with 10% compounding interest paid annually. Calculate the future value.	05	2	4

(P.T.O.)



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of Civil Engineering
In-Semester Examination / Evaluation I
Academic Year: 2025-26 (semester I)

Class: Final Year B.Tech. Course: Water Conservation and Management Course Code: PEC-CE407

Date: 18.08.2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.			Marks	CO	BT
Q.1	a	Illustrate in detail mechanism of soil erosion	06	CO1	BT2
	b	Describe in detail soil erosion control structures.	04	CO1	BT2
Q.2	a	Design cement nala bandhara with following given data (i) Catchment area = 90 ha (ii) Storage capacity = 4.5 TMC (iii) High Flood level (HFL) = 102 m (iv) Full tank level (FTL) = 101.5 m (v) Nala Bed Level (NBL) = 100 m (vi) Foundation bed level = 98 m (vii) Weir bottom R. L. = 98.5 m (viii) Storage height = 1.5 m (ix) Free Board = 0.4 m (x) Batter = 1:0.3	05	CO3	BT4
	b	Classify and explain soil and water conservation measures	03	CO2	BT3
	c	Describe water harvesting techniques.	02	CO2	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

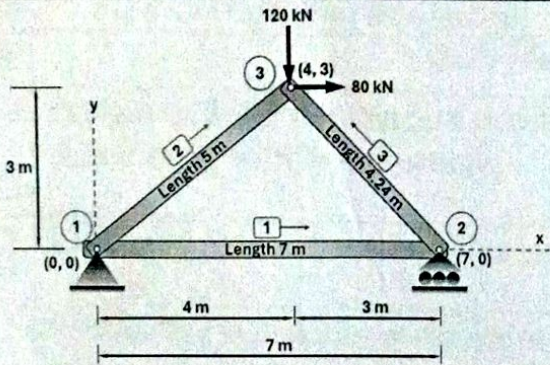
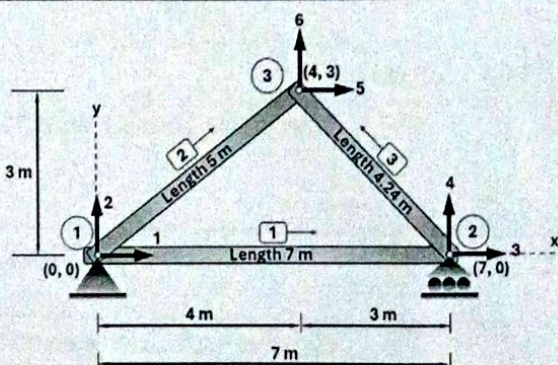
Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Recall the concepts of WRE-I & II
CO 2	Understand the concepts related to Soil and water conservation techniques and watershed management
CO 3	Apply the knowledge of theories and equations to drainage design, irrigation system management, etc.
CO 4	Analyse the different practice for soil and water conservation.

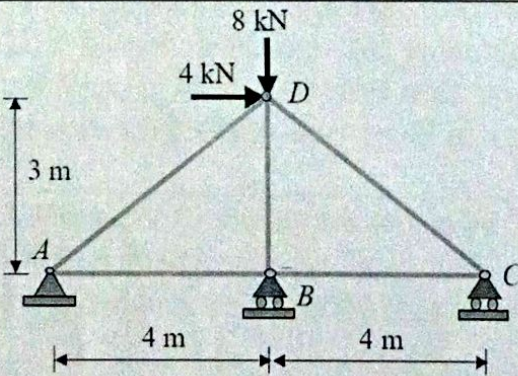
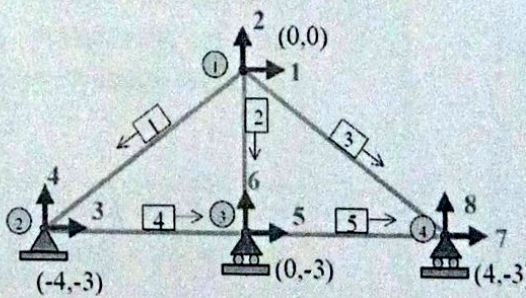


Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.
- vi. Solve either question 1 or question 2 (Any one)

Q.N.	Statement	Marks	CO	BT
Q.1	For the truss shown in Figure 1 (a), use the stiffness method to; i) Determine the deflection of the loaded joint ii) Determine the end forces of each member and the reactions at support Take $AE = 8 \times 10^3 \text{ kN}$, refer the node numbering, node coordinates, member numbering, and degrees of freedom as shown in the figure 1 (b)	20	CO 1	BT 4
 Figure 1 (a)		 Figure 1 (b)		

OR

Q.2	For the truss shown in Figure 2 (a), use the stiffness method and write the global stiffness matrix for the entire truss. Write the matrix equation showing the relationship between the forces and the displacements. Take $AE = 8 \times 10^3 \text{ kN}$, refer the node numbering, node coordinates, member numbering, and degrees of freedom as shown in the figure 2 (b)	20	CO 1	BT 4
 Figure 2 (a)		 Figure 2 (b)		

=== ALL THE BEST ===

Bloom's Taxonomy (BT)- BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

On successful completion of this course, students should be able to:

- CO1 Recall the fundamentals of TOS-I and TOS-II.
- CO2 Understand the basics of direct stiffness method in the analysis of structural components.
- CO3 Apply the different methods of analysis of structural components in practical problems using finite element analysis.
- CO4 Analysis of plane frames, plane grids, plane trusses using basic knowledge of direct stiffness method and fundamentals of finite element analysis.